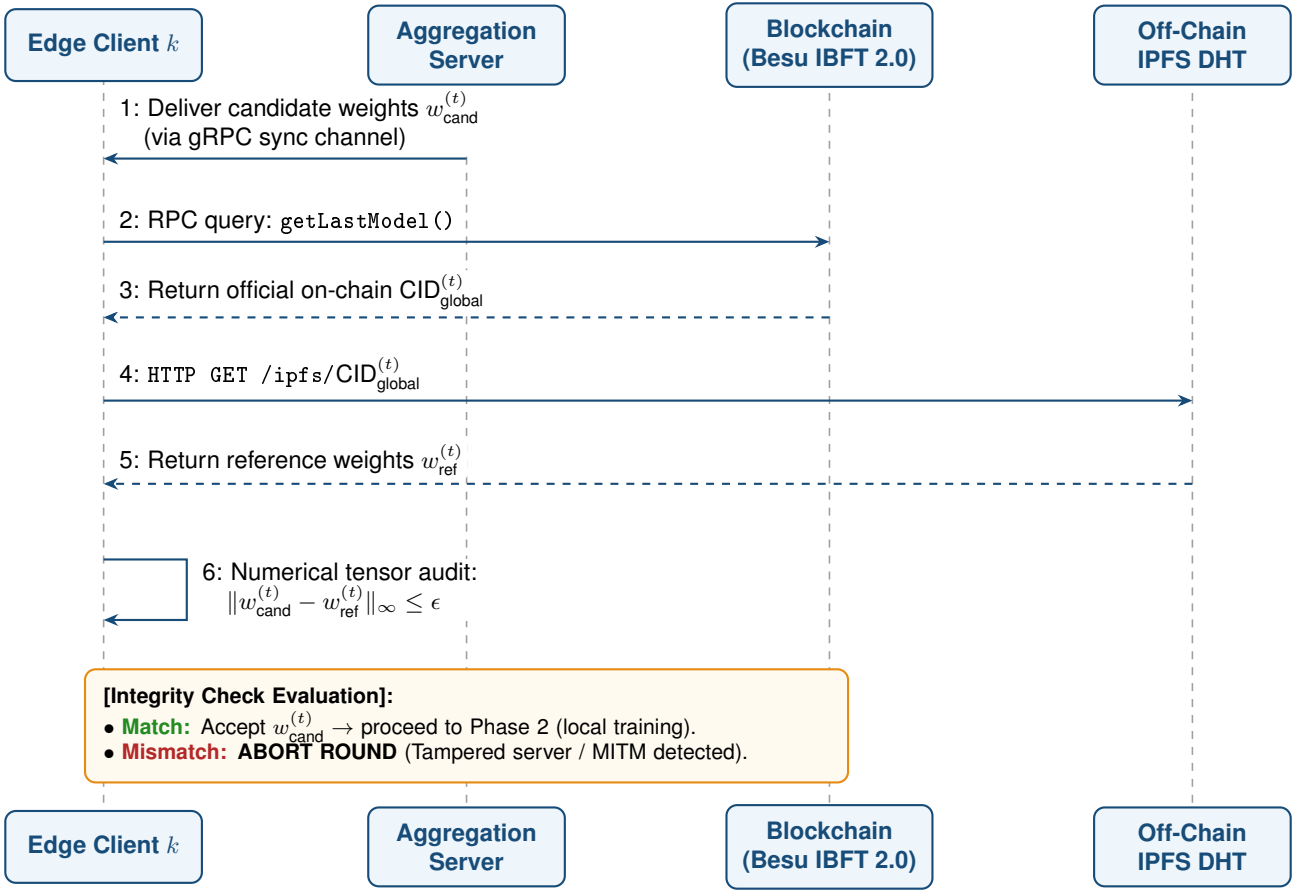

Edge Client k
Aggregation Server
Blockchain (Besu IBFT 2.0)
Off-Chain IPFS DHT
1: Deliver candidate weights $w_{\text{cand}}^{(t)}$ (via gRPC sync channel)
2: RPC query: getLastModel()
3: Return official on-chain $\text{CID}_{\text{global}}^{(t)}$
4: HTTP GET /ipfs/$\text{CID}_{\text{global}}^{(t)}$
5: Return reference weights $w_{\text{ref}}^{(t)}$
6: Numerical tensor audit: $\|w_{\text{cand}}^{(t)} - w_{\text{ref}}^{(t)}\|_\infty \leq \epsilon$
[Integrity Check Evaluation]:
• Match: Accept $w_{\text{cand}}^{(t)} \rightarrow$ proceed to Phase 2 (local training).
• Mismatch: ABORT ROUND (Tampered server / MITM detected).
Edge Client k
Aggregation Server
Blockchain (Besu IBFT 2.0)
Off-Chain IPFS DHT